

WAN Technologies — Guided Notes ANSWER KEY

Unit 1.8.3 | CompTIA Network+ N10-009 Obj. 1.8 | N10-008 1.8

For instructor use / distribute after students complete the notes.

Question 1

Circuit switching establishes a _____ path for the duration of communication, guaranteeing _____ but wasting capacity when idle. Packet switching breaks data into _____ that share carrier infrastructure, which is more _____ but provides no dedicated bandwidth.

Answer: Dedicated; bandwidth/performance; packets; efficient.

Question 2

A T1 leased line provides _____ Mbps. A T3 provides _____ Mbps. Leased lines offer _____ bandwidth and _____ performance. Their main disadvantage compared to modern alternatives is _____.

Answer: 1.544 Mbps; 44.736 Mbps; guaranteed/dedicated; predictable; high cost per megabit.

Question 3

MPLS routes traffic using _____ instead of IP addresses, creating virtual circuits. It supports _____ (two sites) and _____ (all sites can communicate directly) configurations. Its main advantage over broadband is _____ for latency, jitter, and packet loss.

Answer: Labels; point-to-point; any-to-any (full mesh); SLA guarantees.

Question 4

Metro Ethernet extends _____ connectivity beyond a single building using carrier _____. From the router's perspective it looks like _____. It is typically limited to _____ geographic areas.

Answer: Ethernet; metropolitan fiber network; a very long Ethernet cable (same frame format, same interface); metro/city-wide.

Question 5

Match each broadband type to its key characteristic:

Cable → _____

DSL → _____

Fiber → _____

Answer: Cable → shared neighborhood bandwidth, speeds vary at peak hours, asymmetric; DSL → runs over telephone copper, speed degrades with distance from CO, declining technology; Fiber → dedicated to premises, symmetrical speeds up to 1+ Gbps, best performance.

Question 6

An SD-WAN appliance at a branch office monitors _____, _____, and _____ on all WAN links in real time. Application-aware policies send _____ traffic over the highest-quality link and _____ traffic over cheaper broadband.

Answer: Latency; jitter; packet loss (bandwidth); voice/video (latency-sensitive); web browsing/cloud application.

Question 7

The 'last mile' refers to _____. It is often the _____ in WAN performance. Installing fiber last mile to a building that doesn't have it can take _____, which affects WAN planning timelines.

Answer: The connection between the carrier's network infrastructure and the customer's building; bottleneck; weeks to months.

Question 8 — Real World Application

REAL WORLD: Your organization has 20 branch offices each paying \$1,800/month for MPLS. A vendor proposes replacing each with \$150/month fiber broadband plus an SD-WAN appliance. What questions do you ask before approving the change?

Answer: 1. Is fiber available at all 20 locations (last mile)? 2. What are the SLA terms on the fiber broadband vs current MPLS? 3. Which applications are latency/jitter sensitive (voice, video, real-time systems) — will SD-WAN's dynamic path selection meet their requirements? 4. What is the SD-WAN licensing and appliance cost — calculate true TCO not just circuit savings? 5. What is the failover/redundancy plan if the single broadband link goes down (LTE backup)? 6. Does the SD-WAN solution support existing security policies and integrations?